

Gabriela Berenice Díaz Cortés

Researcher in Applied Mathematics & Scientific Computing — Mexican Institute of Petroleum (IMP)
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Profile

Physicist and applied mathematician specialized in computational modeling of multiphase flow in porous media, reservoir simulation, physics-informed neural networks (PINNs), optimization methods and linear solvers. PhD in Applied Mathematics from TU Delft. Supervisor of PhD, MSc and internship students; lecturer of scientific programming with Python.

Work Experience

Researcher — Mexican Institute of Petroleum (IMP)

2018 – Present · Mexico City, Mexico

- Energy transition roadmap (2025–).
- Geological hydrogen and its prospective potential in Mexico (2025–).
- Artificial intelligence techniques applied to the simulation of fluid dynamics (2023–2025).
- AI-based simulator prototype to quantify water coning effects in wells with Venturi flow pattern enhancer systems (2023–2025).
- Thermodynamic characterization of heavy oils by the method of functional groups (2022–2024).
- Near-wellbore simulation of naturally fractured carbonate reservoirs integrating proxy models (2020–2022).
- Supercomputing applied to the exploration and production of hydrocarbons (2018–2020).

Visiting Professor, Artificial Intelligence Laboratory — Computing Research Center, National Polytechnic Institute (CIC-IPN)

2021 – 2024 · Mexico City, Mexico

- Teaching: Mathematical Methods, Linear Programming.
- Project: AI techniques for the optimization of vehicular traffic modeling (2020–2021).

Education

PhD in Applied Mathematics — Delft Institute of Applied Mathematics, TU Delft, The Netherlands

2014 – 2019

- Thesis: *POD-based deflation method for reservoir simulation*. Supervisors: Dr. Cornelis Vuik and Dr. Jan Dirk Jansen.

MSc in Material Sciences — Instituto de Investigaciones en Materiales, UNAM, Mexico

2011 – 2014 · Grade 9.2, *Cum Laude*

- Thesis: *Dynamic Speckle for Materials Characterization*. Supervisors: Dr. Tupak E. García Fernández, Dr. Mayo Villagrán Muniz, Dr. Francisco M. Sánchez Arévalo.

BSc in Physics (Optics) — Facultad de Ciencias, UNAM, Mexico

2005 – 2010 · Grade 8.7

- Thesis: *Laser ablation of NiMnSn and NiMnIn alloys for obtaining thin films*. Supervisor: Dr. Tupak E. García Fernández.

Journal Articles

- J. E. González, **G. B. Díaz-Cortés**, R. M. del Castillo, R. Espejel, A. Calles (2025). Evolutionary algorithms applied to minimize the thermal transport in atomic nanoribbons. *Computational Materials Science*, 260, 114195.
- Z. Sánchez-López, **G. B. Díaz Cortés** (2025). Optimizing a Physics-Informed Neural Network (PINN) to solve the Reynolds Equation. *Revista Mexicana de Física*, 71(2), 020601.
- **G. B. Díaz-Cortés**, R. Luna-García (2023). A Novel Evolutionary Algorithm: One-Dimensional Subspaces Optimization Algorithm (1D-SOA). *Symmetry*, 15(10), 1873.

- **G. B. Díaz Cortés**, C. Vuik, J. D. Jansen (2021). Accelerating the solution of linear systems appearing in two-phase reservoir simulation by the use of POD-based deflation methods. *Computational Geosciences*, 25.
- **G. B. Díaz-Cortés**, C. Vuik, J. D. Jansen (2018). On POD-based Deflation Vectors for DPCG applied to porous media problems. *Journal of Computational and Applied Mathematics*, 330, 193–213.
- M. A. Valverde-Alva, T. García-Fernández, **G. Díaz-Cortés** et al. (2014). Expanded use of a fast photography technique to characterize laser-induced plasma plumes. *Revista Mexicana de Física*, 60(3), 195–204.

Conference Proceedings

- Z. Sánchez, **G. Díaz-Cortés**, G. Domínguez-Zacarías, G. Fuentes-Cruz (2025). Modeling the Diffusion Equation with PINNs and ANNs. *Intl. Congress of Computer Science (ACSAR)*, Springer.
- E. Cabrera-Navarrete, **G. Díaz-Cortés** (2025). Physics-Informed Neural Networks for Approximating the Solution of the 1D Poisson Equation. *Intl. Congress of Computer Science (ACSAR)*, Springer.
- L. Sheremetov, L. A. López-Peña, **G. B. Díaz-Cortés**, D. A. López-Falcón, E. E. Luna-Rojero (2023). Deep Learning Model of Two-Phase Fluid Transport Through Fractured Media: A Real-World Case Study. *LOD 2023*, Springer, 55–68.
- **G. B. Díaz Cortés**, J. D. Jansen, C. Vuik (2018). On the Acceleration of Ill-Conditioned Linear Systems: A POD-based Deflation Method for the Simulation of Two-Phase Flow. *ECMOR XVI*, EAGE.
- **G. B. Díaz Cortés**, C. Vuik, J. D. Jansen (2016). Physics-based Pre-conditioners for Large-scale Subsurface Flow Simulation. *ECMOR XV*, EAGE.

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- Two-Phase Flow Simulator in Cylindrical Coordinates for Analyzing the Effect of a Venturi Flow Pattern Enhancer on Water Coning (2025). Reg. No. 03-2025-071810114400-01.
- Molar Gibbs Free Energy Optimization Algorithm with Linear Constraints for Determining Chemical Pseudo-Structures of Petroleum Fractions in Python (2024).
- Computational simulation of the acid stimulation of an oil well: mathematical model, numerical implementation and validation with well test data (2024).
- Two-phase reservoir simulation near wellbore in cylindrical coordinates with petrophysical information and acid injection (2021–2022, series of registrations).

Supervision & Tutoring

- PhD (main tutor): *AI techniques applied to the simulation of fluid dynamics*, Zoila Sánchez López — IMP, graduated 2025.
- PhD (main tutor): *Physics-Informed Machine Learning Applied to Fluid Flow in Porous Media*, Emmanuel A. Cabrera Navarrete — IMP, 2024–.
- PhD (main tutor): *Optimization of the Operating Window through Mathematical Modeling and AI*, Patricia Domínguez Ramírez — IMP, 2025–.
- PhD committee: *AI models applied to hydrogen production from biomass gasification*, Sindy D. Puerto Vargas — IMP, ongoing.
- MSc (main tutor): *Deflation techniques and POD methods for the acceleration of Krylov subspace methods*, Jenny Tjan — TU Delft, 2018.
- Internships: reservoir simulation, machine learning for flow units, bibliometric databases, AI for high-energy astrophysics (IMP and CBPF Brazil, 2022–ongoing).

Teaching

- Basic Python Programming — IMP postgraduate program (2023, 2025, 2026).
- Advanced Python Programming — IMP postgraduate program (2026).
- Research Seminar — IMP postgraduate program (2023–2025, spring and autumn).
- Introduction to Scientific Computing with Python — IMP (2023).
- Mathematical Methods and Linear Programming — CIC-IPN (2021).
- Teaching assistant: Computational Physics (UNAM, 2025); Numerical Mathematics and Differential Equations (TU Delft, 2014–2017); Electromagnetism Lab (UNAM, 2010–2014).

Conference Participation

- CORE International Conference on Computer Science (CORE-CIC), IPN, Mexico (2025) — two presentations on PINNs for the diffusion and 1D Poisson equations.
- XXIX Conference of the Fluid Dynamics Division, Mexican Physics Society, Morelia (2023) — talks on water coning simulation, acid stimulation and PINNs for fluid flow.
- 9th Intl. Conference on Machine Learning, Optimization, and Data Science (LOD), UK (2023) — talk on deep learning for two-phase transport in fractured media.
- Latin American Conference on Industrial and Applied Mathematics (LACIAM), Rio de Janeiro (2023) — poster on metaheuristic optimization for astrophysical blazar modeling.
- SIAM Conference on Mathematical & Computational Issues in the Geosciences (GS21), virtual (2021) — talk on POD-based deflation for two-phase reservoir simulation.
- Latin America High Performance Computing Conference (CARLA), virtual (2021) — poster on AI for vehicular flow optimization.
- European Conference on the Mathematics of Oil Recovery (ECMOR), EAGE — Amsterdam (2016) and Barcelona (2018), talks on preconditioners and POD-based deflation.
- SIAM Geosciences (GS17), Erlangen, Germany (2017) — minisymposium organizer, *Krylov-based Deflation Methods in Flow for Porous Media*, and talk.
- Intl. Conference on Preconditioning Techniques — Eindhoven (2015) and Vancouver (2017, talk).
- Woudschoten Conferences on Scientific Computing, Zeist, The Netherlands (39th–43rd, 2014–2018).

Certifications & Continuing Education

- Claude 101, Claude Code 101, Claude Code in Action, Introduction to Claude Cowork — Anthropic Academy (2026).
- Advanced Scientific Programming in Python — Latin America, UNAM (2023).
- Introduction to Machine Learning — Coursera, Duke University (2021); A Crash Course in Data Science — Coursera, Johns Hopkins University (2021); Introduction to Data Science in Python — Coursera, University of Michigan (2021).
- Fundamentals and Applications of Data Science — IMP (2021).
- Petrel Structural Modeling and ECLIPSE Advanced Topics — NExT, a Schlumberger company (2019).
- Machine Learning for Industry — CIC-IPN (2019).

Awards & Scholarships

- Award for the use of the National Supercomputing Laboratory of Southeast Mexico — project *Machine Learning techniques applied to the modeling of physical systems* (2022).
- SIAM Student Chapter Certificate of Recognition, TU Delft (2017–2018).
- PhD scholarship CONACYT-SENER/IMP (2014–2018); MSc scholarship CONACYT (2011–2013).
- National Standard of Competency EC0217 — in-person group training (CONOCER, 2019).

Skills & Languages

- Programming: Python, MATLAB, C/C++, MPI/OpenMP.
- Software: Linux, Windows, LaTeX, Mathematica, Petrel, Eclipse, Origin, Weka.
- Languages: Spanish (native), English (fluent), French (advanced), German and Dutch (basic).